Module 8

# Risk, Return, and the CAPM

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

# Goals

- **Module 7** left two questions open,
  - What is risk?
  - How is the opportunity cost of an investment (which we use as the discount rate) related to risk?

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  - How is the opportunity cost of an investment (which we use as the discount rate) related to risk?
- This module answers both,
  - **Part 1**, returns and risk premiums
  - **Part 2**, return volatility and portfolio diversification
  - **Part 3**, beta, the risk that actually matters
  - **Part 4**, CAPM, the relationship between beta and expected return
  - **Part 5**, from expected return to discount rate

## Module 8, Part 1

# Returns and Risk Premiums

# Return

Recall from **Module 4**

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- Suppose Oracle's stock price was \$100 at the beginning of 2024 and \$180 at the end of 2024. What was the return for 2024?

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- Suppose Oracle's stock price was \$100 at the beginning of 2024 and \$180 at the end of 2024. What was the return for 2024?

$$\text{Return} = \frac{\$180 - \$100}{\$100} = 80\%$$

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- E.g., a 50% chance of a 10% return and a 50% chance of a 0% return give $E[R_i] = 0.5 \times 10\% + 0.5 \times 0\% = 5\%$

# Example, Expected Returns

- You want the expected return on a stock over the next year. Three things might happen,
  - A recession, 20% chance, the stock returns $-10\%$
  - A normal year, 50% chance, the stock returns 5%
  - A boom, 30% chance, the stock returns 15%

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  - A normal year, 50% chance, the stock returns 5%
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- Solution, the expected return is

$$E[R_i] = 0.2 \times (-10\%) + 0.5 \times 5\% + 0.3 \times 15\% = 5\%$$

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Its stock portfolio on June 30, 2026 ($20.2 billion),

| Company | Business | Weight |
|---|---|---|
| Sandisk | Memory chips | 28% |
| Micron | Memory chips | 28% |
| Bloom Energy | Fuel cells | 9% |
| TSMC | Chip maker | 6% |
| Nebius | AI cloud | 6% |
| CoreWeave | AI cloud | 4% |
| Other holdings | | 19% |
| Total | | 100% |

Source, SEC Form 13F, filed August 14, 2026

# Portfolio Returns

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$$R_P = w_1 \times R_1 + w_2 \times R_2 + \cdots + w_S \times R_S$$

- The expected portfolio return is likewise a weighted average of the securities' expected returns,

$$E[R_P] = w_1 \times E[R_1] + w_2 \times E[R_2] + \cdots + w_S \times E[R_S]$$

# Example, Portfolio Returns

- One year ago, you bought two stocks for \$500 each, \$1,000 in total
- One stock returned 30% during the year, the other $-10\%$
- What is your portfolio return over the year?

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- Solution, you invested $500/1{,}000 = 50\%$ in each stock, so the weights are 50% each,

$$R_P = w_1 \times R_1 + w_2 \times R_2 = 0.5 \times 30\% + 0.5 \times (-10\%) = 10\%$$

# Example, Expected Portfolio Returns

- You want the expected return on your portfolio over the next year. You hold two stocks, 50% of your money in each
- The first stock's expected return is 5%, the second's is 9%
- What is your portfolio's expected return over the next year?

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- Solution, the expected portfolio return is

$$E[R_P] = w_1 \times E[R_1] + w_2 \times E[R_2] = 0.5 \times 5\% + 0.5 \times 9\% = 7\%$$

# The Risk-Free Rate and Market Returns

- $r_{f,t}$, the **risk-free rate** as of time $t$
  - The return from an investment with no risk, i.e., return is the same in every scenario
  - In practice, the interest rate on Treasury bills (short-term U.S. government debt)

# The Risk-Free Rate and Market Returns

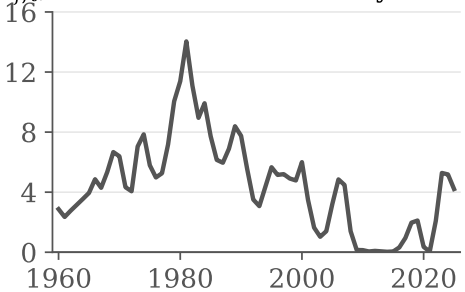
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- $R_{\text{mkt},t}$, the return on the market portfolio in period $t$
  - **Market portfolio**, the (theoretical) portfolio of all investable assets in the economy, weighted by market capitalization
  - In practice, represented by a broad market index such as the S&P 500

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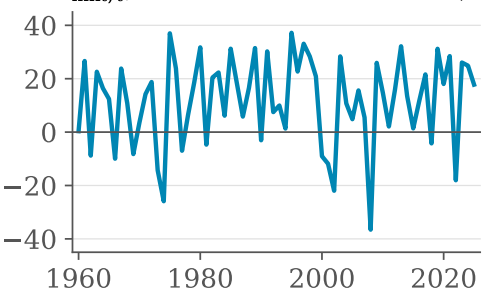
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- $E[R_{\text{mkt}}]$, the **expected return** on the market portfolio

# The Risk-Free Rate and Market Returns over Time

$r_{f,t}$, interest rate on Treasury bills (%)

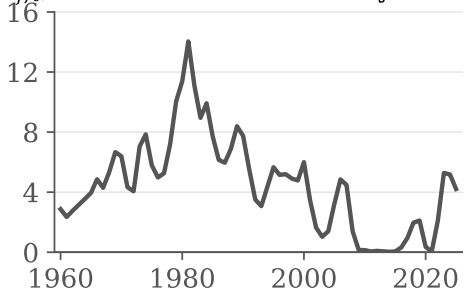



$R_{\text{mkt},t}$, return on the S&P 500 (%)

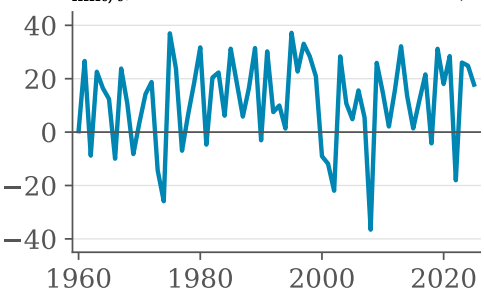


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$r_{f,t}$, interest rate on Treasury bills (%)




$R_{\text{mkt},t}$, return on the S&P 500 (%)




- $r_{f,t}$ and $R_{\text{mkt},t}$ can be different at different $t$

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| Portfolio | Average annual return 1928–2025 | Average extra return over Treasury bills |
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| Stocks | 11.9% | 8.5% |

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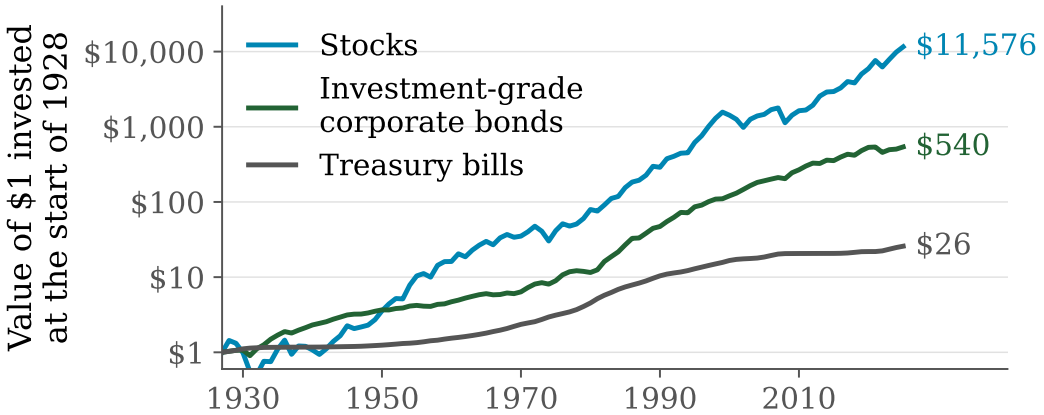
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- The average extra return over Treasury bills is our estimate of the risk premium
- For investment-grade corporate bonds that is 3.5%, for stocks 8.5%

# Returns over Time for Stocks, Bonds, and Treasury Bills



# Recap of Part 1

- **Return**, percentage growth of an asset's value over some period of time

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- $E[R_i] = r_f + \text{Risk premium}_i$, riskier assets have higher expected returns
- The risk premium of stocks is about 8.5%

## Module 8, Part 2

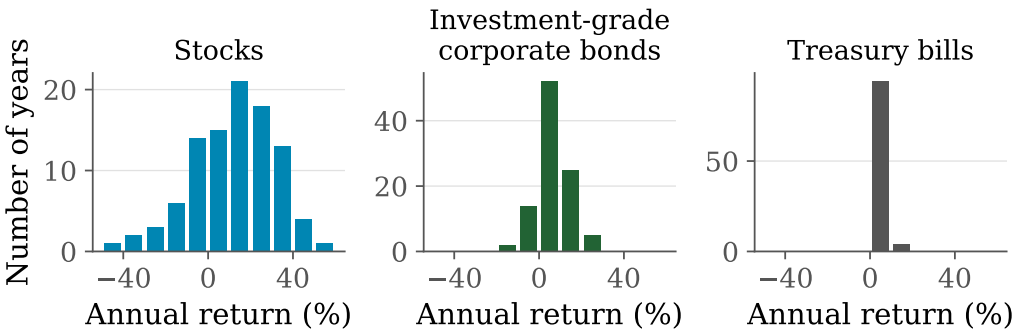
# Return Volatility and Portfolio Diversification

# The Risk Premium of Stocks Is High, but What about Volatility?

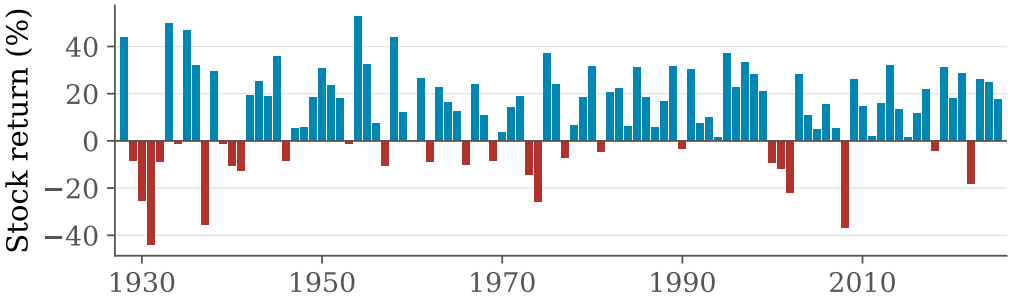
- How volatile are portfolios of stocks vs. bonds vs. Treasury bills?

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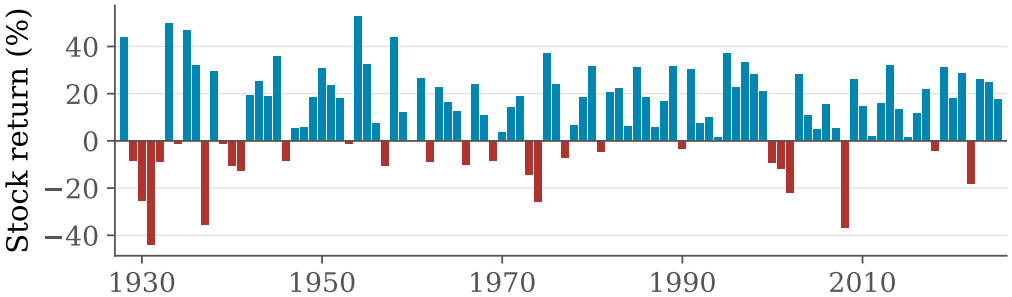
- How volatile are portfolios of stocks vs. bonds vs. Treasury bills?
- Consider the variation in annual returns, 1928–2025, how many years fell in each range,



# Stock Market Returns over Time

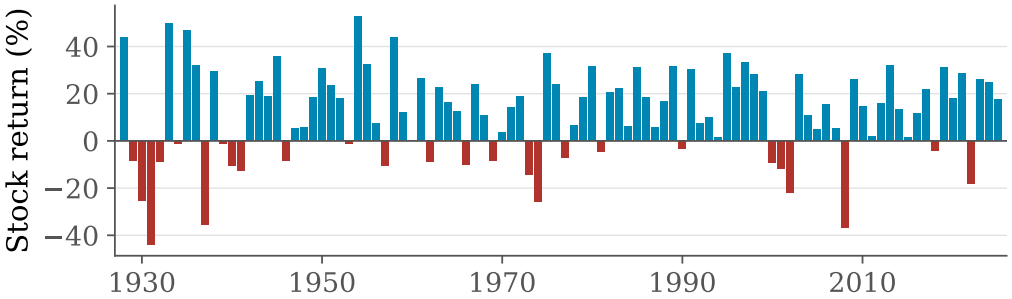


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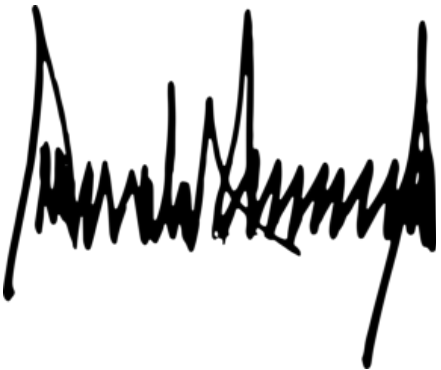
# Stock Market Returns over Time




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- Individual stocks are even more volatile, around 50% of stock-years have negative returns

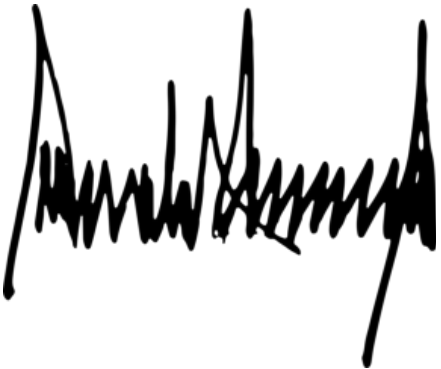
# Discuss, Do They Look Alike?

**US President Donald Trump's Signature**

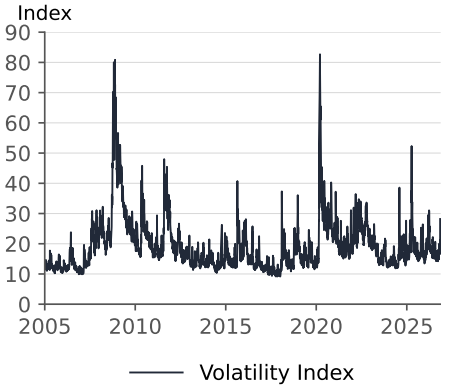


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**US Stock Volatility Index (VIX)**



# Measuring Volatility, Variance and Standard Deviation

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- **Standard deviation**, the square root of the variance,

$$\text{Standard deviation} = \sqrt{\text{Variance}}$$

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| 5 | 5% |
| 6 | 20% |

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| Year | Return | Deviation from average |
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| 1 | 5% | −5% |
| 2 | 25% | 15% |
| 3 | 10% | 0% |
| 4 | −5% | −15% |
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| Year | Return | Deviation from average | Squared deviation |
|---|---|---|---|
| 1 | 5% | −5% | 0.0025 |
| 2 | 25% | 15% | 0.0225 |
| 3 | 10% | 0% | 0.0000 |
| 4 | −5% | −15% | 0.0225 |
| 5 | 5% | −5% | 0.0025 |
| 6 | 20% | 10% | 0.0100 |
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| 6 | 20% | 10% | 0.0100 |
| Average | 10% | Variance (average of squared deviations) | 0.0100 |

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| | | Standard deviation (square root) | 10% |

# Standard Deviation of Treasury Bills vs. Bonds vs. Stocks

| Portfolio | Standard deviation (%), 1928–2025 |
|---|---|
| Treasury bills | 3.0 |

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| Stocks | 19.4 |

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| Stock | Since | Standard deviation of annual returns (%) |
|---|---|---|
| Tesla | 2011 | 201.4 |
| Amazon | 1998 | 186.9 |
| Apple | 1981 | 65.1 |
| Microsoft | 1987 | 41.9 |
| McDonald's | 1967 | 41.3 |

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| **S&P 500** | **1928** | **19.4** |

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- This is where diversification will help!

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- Next year, one of three scenarios happens with equal likelihood, a recession, normal times, or a boom
- You consider two stocks, stock A (does relatively better in a boom) and stock B (does relatively better in a recession),

| Scenario | Probability | Stock A (%) | Stock B (%) |
|---|---|---|---|
| Recession | 1/3 | −8 | +20 |
| Normal | 1/3 | +5 | +3 |
| Boom | 1/3 | +18 | −20 |

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|---|---|
| Recession | $20\%$ |
| Normal | $3\%$ |
| Boom | $-20\%$ |

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| Boom | $18\%$ |
| $E[R_A]$ | $5\%$ |

| Stock B | Return |
|---|---|
| Recession | $20\%$ |
| Normal | $3\%$ |
| Boom | $-20\%$ |
| $E[R_B]$ | $1\%$ |

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| Stock A | Return | Deviation from $E[R_A]$ | |
|---|---|---|---|
| Recession | $-8\%$ | $-13\%$ | |
| Normal | $5\%$ | $0\%$ | |
| Boom | $18\%$ | $13\%$ | |
| $E[R_A]$ | $5\%$ | | |

| Stock B | Return | Deviation from $E[R_B]$ | |
|---|---|---|---|
| Recession | $20\%$ | $19\%$ | |
| Normal | $3\%$ | $2\%$ | |
| Boom | $-20\%$ | $-21\%$ | |
| $E[R_B]$ | $1\%$ | | |

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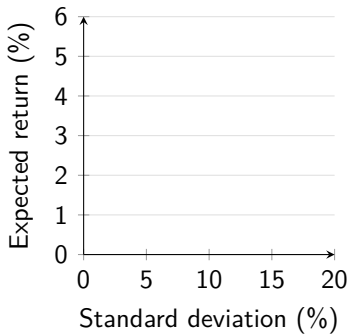
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- Stock A looks better as a stand-alone investment on this metric

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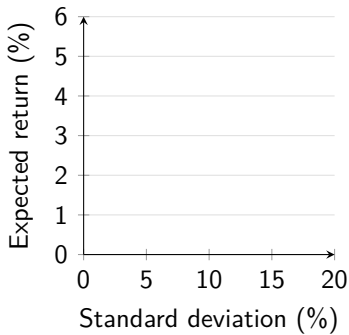
| Portfolio | |
|---|---|
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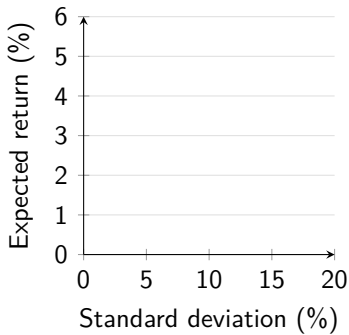
| Portfolio | 1 |
|---|---|
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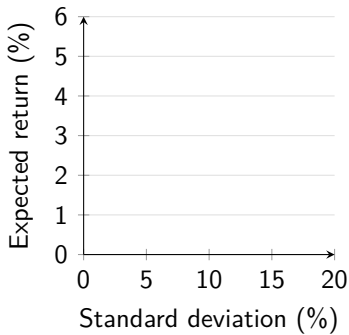
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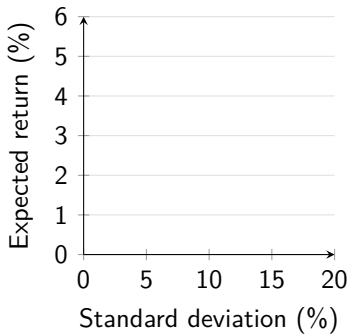
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| Normal | |
| Boom | |
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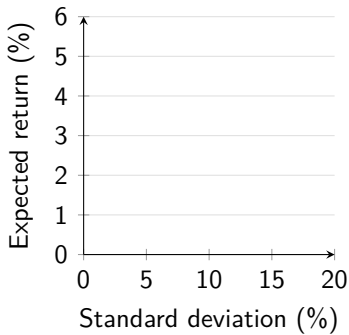
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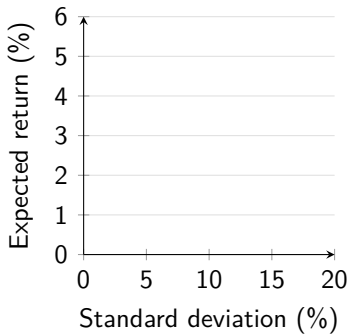
| Portfolio | 1 |
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| Std. dev. | |
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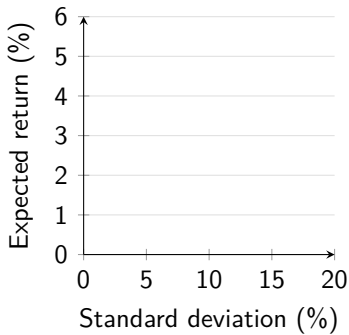
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| Std. dev. | |
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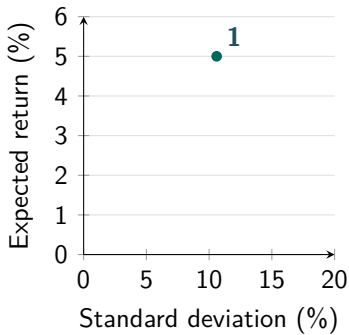
| Portfolio | 1 |
|---|---|
| Stock A weight | 100% |
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| Recession | −8% |
| Normal | 5% |
| Boom | 18% |
| $E[R]$ | 5.0% |
| Std. dev. | 10.6% |
| $E[R]$ / std. dev. | |



# Example, Portfolio Diversification

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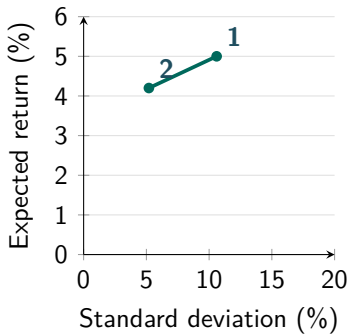
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| $E[R]$ / std. dev. | 0.47 |



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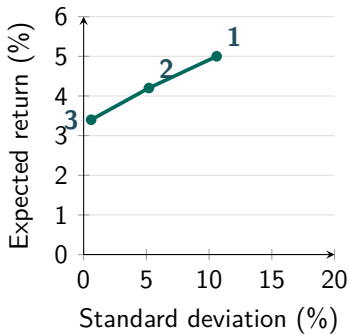
| Portfolio | 1 | 2 |
|---|---|---|
| Stock A weight | 100% | 80% |
| Stock B weight | 0% | 20% |
| Recession | −8% | −2% |
| Normal | 5% | 5% |
| Boom | 18% | 10% |
| $E[R]$ | 5.0% | 4.2% |
| Std. dev. | 10.6% | 5.2% |
| $E[R]$ / std. dev. | 0.47 | 0.80 |



# Example, Portfolio Diversification

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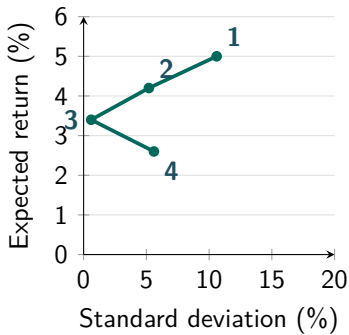
| Portfolio | 1 | 2 | 3 |
|---|---|---|---|
| Stock A weight | 100% | 80% | 60% |
| Stock B weight | 0% | 20% | 40% |
| Recession | −8% | −2% | 3% |
| Normal | 5% | 5% | 4% |
| Boom | 18% | 10% | 3% |
| $E[R]$ | 5.0% | 4.2% | 3.4% |
| Std. dev. | 10.6% | 5.2% | 0.6% |
| $E[R]$ / std. dev. | 0.47 | 0.80 | 5.77 |



# Example, Portfolio Diversification

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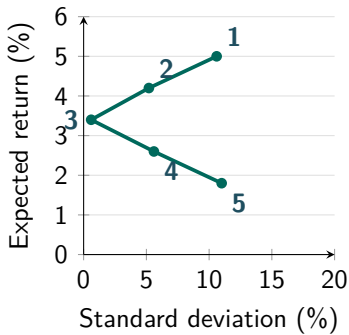
| Portfolio | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Stock A weight | 100% | 80% | 60% | 40% |
| Stock B weight | 0% | 20% | 40% | 60% |
| Recession | −8% | −2% | 3% | 9% |
| Normal | 5% | 5% | 4% | 4% |
| Boom | 18% | 10% | 3% | −5% |
| $E[R]$ | 5.0% | 4.2% | 3.4% | 2.6% |
| Std. dev. | 10.6% | 5.2% | 0.6% | 5.6% |
| $E[R]$ / std. dev. | 0.47 | 0.80 | 5.77 | 0.46 |



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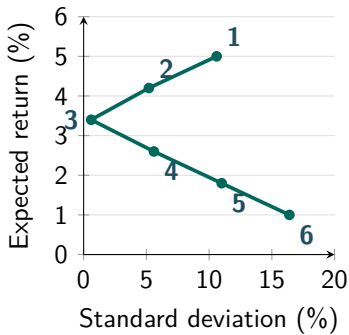
| Portfolio | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Stock A weight | 100% | 80% | 60% | 40% | 20% |
| Stock B weight | 0% | 20% | 40% | 60% | 80% |
| Recession | −8% | −2% | 3% | 9% | 14% |
| Normal | 5% | 5% | 4% | 4% | 3% |
| Boom | 18% | 10% | 3% | −5% | −12% |
| $E[R]$ | 5.0% | 4.2% | 3.4% | 2.6% | 1.8% |
| Std. dev. | 10.6% | 5.2% | 0.6% | 5.6% | 11.0% |
| $E[R]$ / std. dev. | 0.47 | 0.80 | 5.77 | 0.46 | 0.16 |



# Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

| Portfolio | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Stock A weight | 100% | 80% | 60% | 40% | 20% | 0% |
| Stock B weight | 0% | 20% | 40% | 60% | 80% | 100% |
| Recession | −8% | −2% | 3% | 9% | 14% | 20% |
| Normal | 5% | 5% | 4% | 4% | 3% | 3% |
| Boom | 18% | 10% | 3% | −5% | −12% | −20% |
| $E[R]$ | 5.0% | 4.2% | 3.4% | 2.6% | 1.8% | 1.0% |
| Std. dev. | 10.6% | 5.2% | 0.6% | 5.6% | 11.0% | 16.4% |
| $E[R]$ / std. dev. | 0.47 | 0.80 | 5.77 | 0.46 | 0.16 | 0.06 |



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- What is the standard deviation of the portfolio?

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- **Diversification**, spreading the money in a portfolio across many securities rather than one
- Diversification typically makes the standard deviation of a portfolio lower than the weighted average of the standard deviations of the stocks in it,

$$\textbf{StdDev}[R_P] < w_A \,\text{StdDev}[R_A] + w_B \,\text{StdDev}[R_B]$$

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- An economy with $N$ companies, each producing a product

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- An economy with $N$ companies, each producing a product
- Each company's stock returns 20% in a normal year
- Two kinds of risk,
  - A company's product fails (30% chance each year), its stock return falls by 40 points
  - Something bad hits the whole economy, like a pandemic (10% chance each year), every stock return falls by 30 points

# Simulating an Investor's Return

- Simulate many years of this economy, the AI writes the code and plots

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- An investor puts all the money in one company. What is the standard deviation of the investor's return?
- Now the investor splits the money equally across 2 companies, then 5, 10, 30, 100. Does the standard deviation fall? To zero?

# What Is the End Game of Portfolio Diversification?

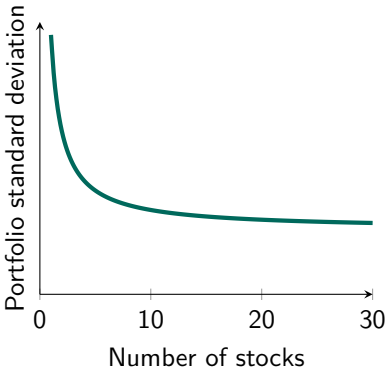
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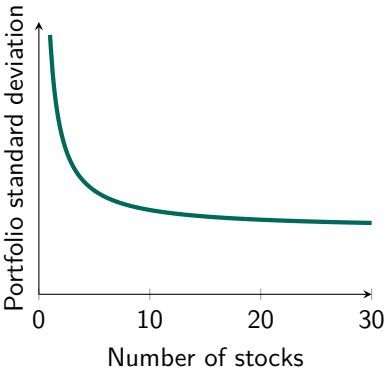
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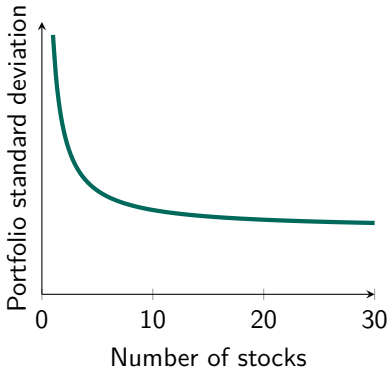
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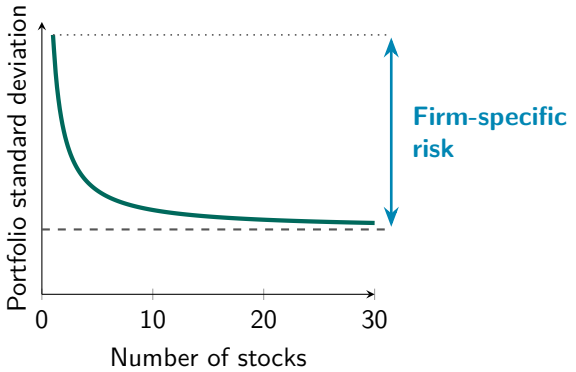
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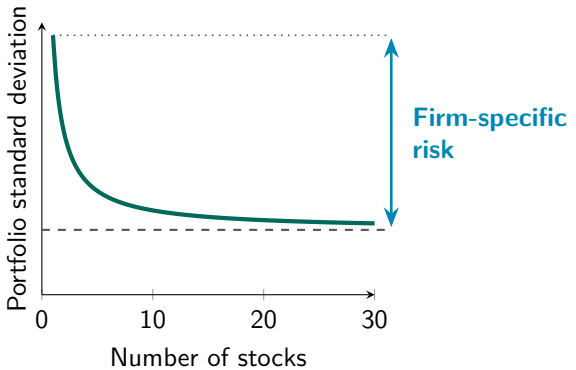
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- Such events average out across many stocks
- Diversification removes firm-specific risk



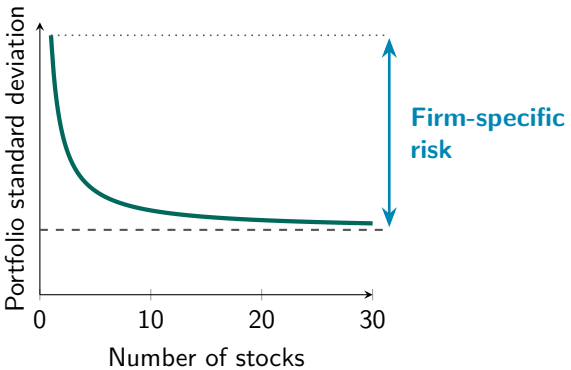
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- Yet there is a limit, the curve never reaches zero



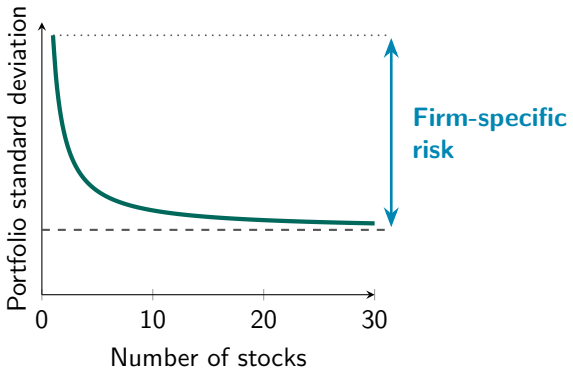
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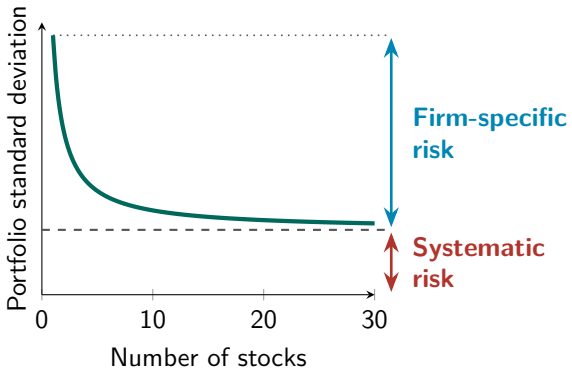
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- Yet there is a limit, the curve never reaches zero
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- Every stock is hit at once, nothing averages out



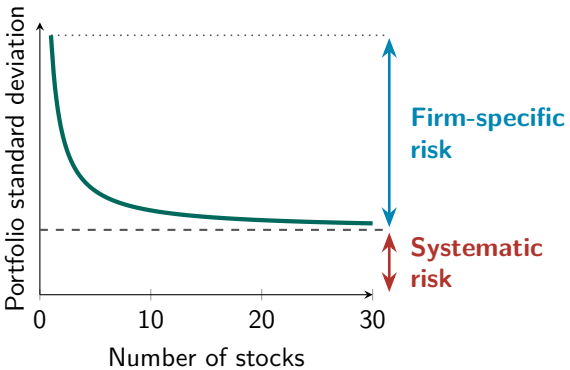
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## Module 8, Part 3

# Beta

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- So the risk of a stock that matters is its systematic risk, not its standard deviation
- How much systematic risk does a stock carry?

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- In practice, represented by a broad market index such as the S&P 500

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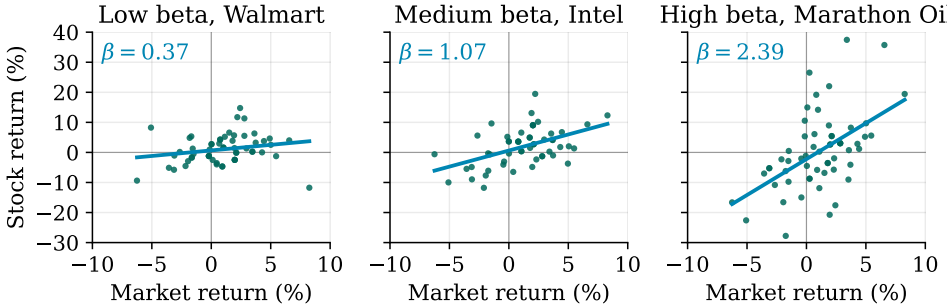
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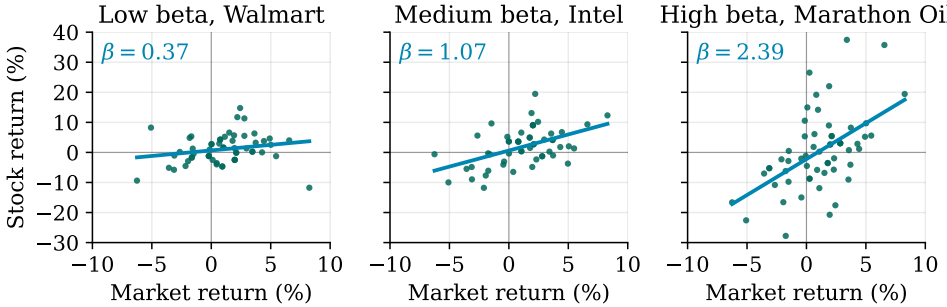
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# Example, Low vs. Medium vs. High Beta Stocks

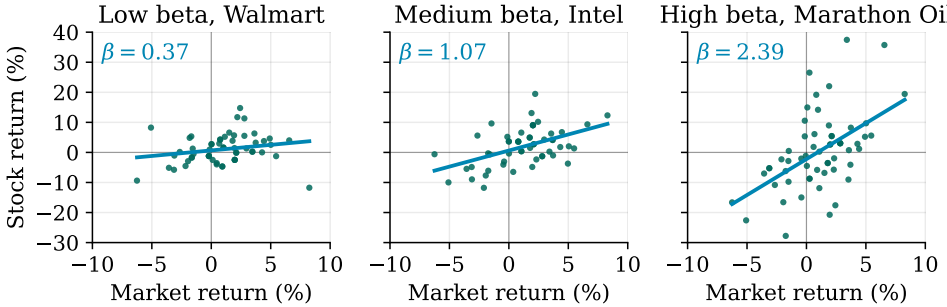


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- Cyclical businesses, whose revenues and earnings depend strongly on the state of the economy, tend to have high betas
- Businesses that produce essentials (food, beer, cosmetics) are less affected by the state of the economy and tend to have low betas

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  - Its return moves one for one with the market's, because it is the market

# A Primer on Regression Analysis

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- **Regression analysis**, a statistical tool for describing the "best-fit" linear relation between two variables, $X$ (the "explanatory" variable) and $Y$ (the "dependent" variable)

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- The best-fit line keeps the residuals $e_i$ as small as possible overall

# Example, Height and Weight

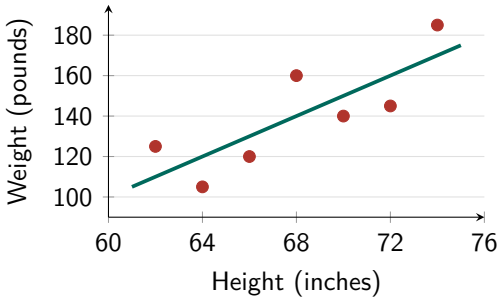
- Seven people's heights and weights, what is the best-fit line between weight ($Y$) and height ($X$)?

| Person | Height (inches) | Weight (pounds) |
|---|---|---|
| 1 | 62 | 125 |
| 2 | 64 | 105 |
| 3 | 66 | 120 |
| 4 | 68 | 160 |
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- Solution, the best-fit line is Weight $= -200 + 5 \times$ Height, so $\alpha = -200$ and $\beta = 5$

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- The coefficient $\beta$ from this regression is the estimate of the stock's beta

# Example, Measuring Beta

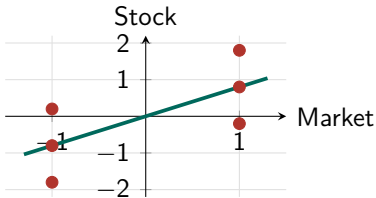
- A company's stock has the following excess returns, alongside the market portfolio's (both in %). What is its beta?

| Month | Market excess return | Stock excess return |
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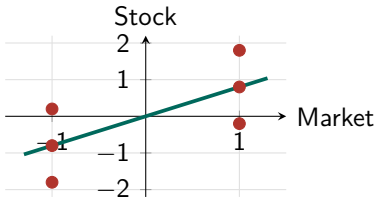



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- The slope is the estimate of the stock's beta, here $\beta = 0.8$

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## Module 8, Part 4

# The CAPM

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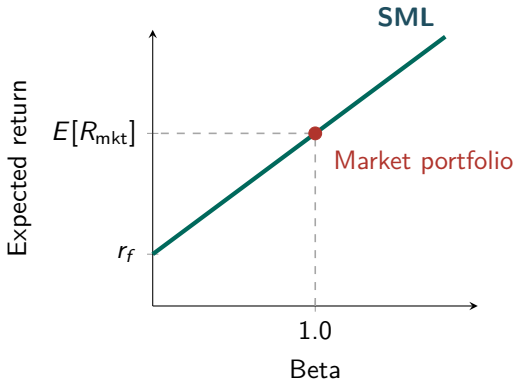
$$E[R_i] = r_f + \beta_i \times (E[R_{\text{mkt}}] - r_f)$$

  - $E[R_{\text{mkt}}] - r_f$, the **market risk premium**, the expected return on the market portfolio over the risk-free rate

# CAPM, the Security Market Line

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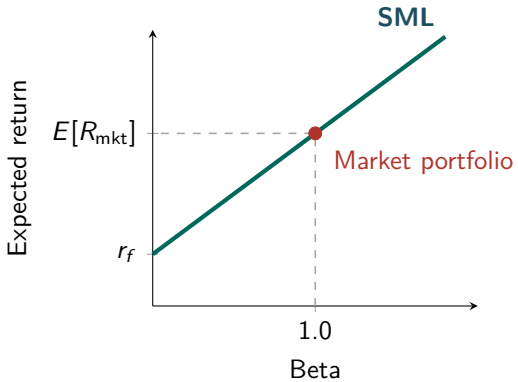


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- A straight line, it starts at $r_f$ for $\beta = 0$ and passes through the market portfolio, $\beta = 1$ and $E[R_{\text{mkt}}]$



# CAPM, Expected Returns for Selected Companies

- With $r_f = 3\%$ and a market risk premium $E[R_{\text{mkt}}] - r_f = 7\%$, the expected returns are,

| Company | Business | Beta |
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| Disney | Entertainment | 1.39 | 12.7 |
| Intel | Chip maker | 1.07 | 10.5 |
| Coca-Cola | Beverages | 0.70 | 7.9 |
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      They require nothing extra for firm-specific risk, because diversification removes it for free

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# Recap of Part 4

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## Module 8, Part 5

# From Expected Return to Discount Rate

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- Is this return the project's discount rate $r$ in capital budgeting?

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- Most companies do borrow, **Module 9** adds debt

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***Discount rate of a project = Expected return on stocks with the same beta***

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3. Discount the project's cash flows at $r$ as in **Module 7**, accept if NPV $> 0$ (equivalently, if IRR $> r$)

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- Same cash flows, opposite answers. The discount rate comes from the project's beta, and accepting only the projects with NPV $> 0$ at this rate makes the shareholders better off

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- Should the oil project be discounted at the company's 6.5%?

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- The company is a portfolio of its two businesses, with weights 80% and 20%, so its beta is the weighted average (**Part 3**),

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- That is exactly the weighted average of each business's own CAPM rate,

$$r_{\text{company}} = 0.8 \times r_{\text{soup}} + 0.2 \times r_{\text{oil}} = 0.8 \times 6.5\% + 0.2 \times 17\% = 8.6\%$$

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- So discount the oil project at 17%, its own rate, not at the company's 6.5%

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- A company can set two or more discount rates for different types of projects

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$$\text{NPV} = 100{,}000 \times \frac{1}{r}\left(1 - \frac{1}{(1+r)^5}\right) - 350{,}000$$

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- Use the project's own beta (the twin method), not the company's

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- Most companies also borrow, so their investors are lenders and shareholders
- **Module 9**, the discount rate becomes the weighted average cost of capital (WACC), a mix of what lenders and shareholders require
- Then debt policy, does the mix of debt and equity change what a company is worth?

# Thank you!

Please let me know if you have any questions.